Automated Main Concept Analysis in Narrative Speech for Detecting Linguistic Impairments

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Introduction: Discourse analysis provides insight into language abilities beyond the sentence level, often assessed through storytelling. Traditional analyses have primarily focused on microlinguistic features, like mean length of utterance and type-token ratio, but not macrolinguistic features, like Main Concept Analysis (MCA) [1], topic coherence, and sequencing, which quantify how well speakers convey story ideas, relate utterances to the topic, and maintain narrative order. Despite their diagnostic value [2], macrolinguistic analysis remains largely manual [3] [4]. We developed an NLP-based pipeline that automatically extracts main concept (MC), coherence, and sequencing features from Cinderella story retellings and then applied machine learning to evaluate the diagnostic potential of automated macrolinguistic features in distinguishing Healthy Controls (HC), people with dementia (PWD), and people with aphasia (PWA).

Methods: Audio samples were drawn from two publicly available English-language discourse corpora: AphasiaBank [5] and DementiaBank [6]. The final sample included HC (n=113), PWA (n=102), and PWD (n=94). Audio recordings were transcribed with Amazon Transcribe, and utterances were segmented by terminators and conjunctions.

We implemented an embedding-based MCA using the Cinderella MC inventory [4], though our approach can operate on any narrative discourse with a reference MC list. MC and participant utterances were embedded with the all-mpnet-base-v2 Sentence-Transformers model [7], and MC embeddings were averaged to form a centroid, representing the semantic core of the story. An utterance was classified as an MC if it was within one SD of the centroid. To evaluate the MC identification algorithm, we compared automated MC classifications with human judgments. Utterances that met this criterion were then matched to the MC in the inventory with the highest cosine similarity. We computed sequence scores [9] and developed related algorithms for generating coherence error measures, as shown in Figure 2 [8]. Overall, we computed 23 macrolinguistic measures, including MC metrics (distance to centroid; unique and total MC counts and ratios), sequence score, local and global coherence error ratios, and multiple coherence error type counts and ratios (topic switching, missing referents, tangential utterances, propositional repetitions, fillers, conceptual incongruences, and coherent utterances). For model training, we retained only ratio-based features to minimize the influence of speech quantity. We conducted a 3-way multinomial logistic regression classification among PWA, PWD, and controls, as well as one-vs-one pairwise classifications, to test if automatically extracted macrolinguistic features could distinguish these groups.

Results: The increasing distances to the centroid from HC to PWD to PWA confirm that the automated MCA captures narrative informativeness across groups (Figure 1). Across all groups, our automated system achieved strong accuracy (84%) in identifying MC utterances when compared with the human-determined labels. Using the reduced set of 10 features by removing highly correlated features $(|r| > .8)$, logistic regression achieved a macro-averaged ROC AUC score of 0.725 for the three-way classification. For PWA vs HC, the model reached a mean AUC of 0.938, indicating outstanding diagnostic performance. For PWD vs HC, the model achieved lower performance with a mean AUC of 0.661, reflecting the milder linguistic deficits of mild cognitive impairment (MCI) participants in the dementia dataset. For PWD vs PWA, the model achieved a mean AUC of 0.575, consistent with substantial overlap in macrolinguistic impairments between PWD and PWA. The distance to centroid feature had very high coefficient magnitudes, demonstrating its importance in distinguishing groups (Figure 3).

Conclusion: Our study introduces a fully automated method to detect differences in macrolinguistic features of speech in PWA, PWD, and HC. While MCA has long been used manually, our findings demonstrate that computational macrolinguistic metrics can capture impairments across populations.

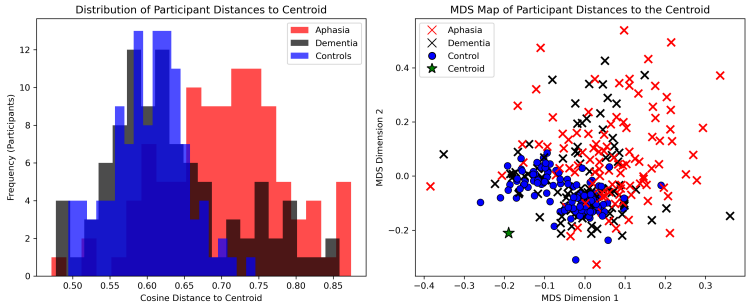

Figure 1: MDS Embedding and Histogram of Participants' Distances to Centroid.

Figure 2: Flowchart for the Coherence Error Algorithm.

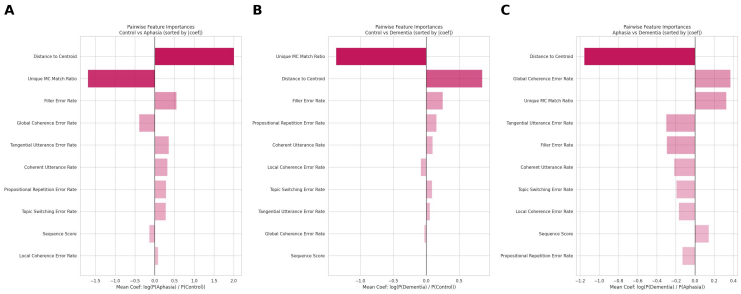

Figure 3: Macrolinguistic Feature Importances for (a) PWA vs HC, (b) PWD vs HC, and (c) PWA vs PWD.

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